COMP-424: Artificial intelligence, Fall 2026

Homework 2

Due on *myCourses* Monday October 19, 9:00pm.

General instructions.

- This is an individual assignment. You can discuss solutions with your classmates, but should only exchange information orally, or else if in writing through the discussion on *Ed* or in an assignment help sessions with a TA. All other forms of written exchange are prohibited. If you did attend an assignment preview session and collaborated with someone (took notes away from your session), please write down the name and email of the person you worked with.
- Unless otherwise mentioned, the only sources you should need to answer these questions are your course notes, the textbook, and the links provided. Any other source used should be acknowledged with proper referencing style (we recommend ACM or IEEE formatting) in your submitted solution.
- For each problem, you can solve manually, or write a program to help you. You can use a programming language of your choice. You can modify code from other sources if you provide adequate citation; this cannot be code from other students in the class (previous or current).
- Submit a single pdf of your written responses through *myCourses*. You can scan-in hand-written pages or typeset the answers. If necessary, learn how to combine many pdf files into one.
- In addition, submit any code developed to answer questions as a separate file through *myCourses*.

Question 1: Constraint satisfaction [50]

Consider the following problem:

You want to solve a simplified 4x4 version of the Sudoku game with the following grid configuration:

2		4	
	1		2
1			3

Each row, column and 2x2 square rooted in one of the 4 corners should contain the digits “1”, “2”, “3”, “4” exactly once.

a) Formulate this problem as a CSP. List the variables, their domains, and the constraints. Draw the constraint graph.

b) Show the first ten steps of backtracking search on this problem, where you order the variables in increasing order first by row, then by column, and the values from lowest to highest. Recall that backtracking search uses a depth-first strategy to expand the search tree.

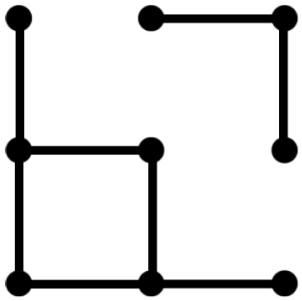
c) Show the first ten steps of backtracking search on this problem *with one-step forward checking*, where you order the variables in increasing order first by row, then by column, and the values from lowest to highest.

d) Write a simple program to implement a *simulated annealing* local search strategy to solve this problem. Submit in writing your problem formulation and document the algorithm design decisions that you make. Run

your algorithm with at least three different sets of design choices (e.g., annealing schedule, initial temperature). Show the first five steps of the algorithm on one run of the program for one setting, including the game state and the number of constraint violations in each step. Run the program 100 times with a maximum of 10000 steps per run. For each of the three settings, report the number of runs in which the program finds a solution, and make note of any other interesting points you observe.

Question 2: Search and Game Playing [30]

You are playing the dots and boxes game on a 3x3 grid shown below. Each player has to draw an edge connecting two dots, if it doesn't already exist. The player who draws the 4th line making a unit square receives a +1 score. The player with the largest score after none of the players can draw edges anymore wins the game.



Assume this is the starting state, and that it's the max player's turn.

a) Apply the Minimax algorithm to the above state, by preferring horizontal moves to vertical moves, and expanding in increasing order first by row, then by column. For convenience, you can represent the moves as (start row, start column, end row, end column), e.g. drawing a vertical line in the bottom right corner can be represented as the move (3, 3, 2, 3) or equivalently (2, 3, 3, 3). Draw the corresponding search tree.
   [Note: the preference for horizontal moves affects how you apply the Minimax algorithm, but not how the players play the game.]

b) Apply the alpha-beta pruning method using the same order of node expansion and show the alpha-beta values for all nodes. Is there any advantage to using alpha-beta pruning?

Question 3: Search Under Uncertainty [20]

a) Play a game of Connections at https://www.nytimes.com/games/connections to familiarize yourself with the rules. (No marks)

b) Formulate the Connections game for a given day as a search problem, formally showing all parts of the search problem as shown in the lectures. Is this an observable, partially observable, or non-observable problem? Deterministic or not?

c) Describe how you could attempt to solve the Connections search problem using an AND-OR search tree. What would the AND and OR nodes represent? How many branches are at each AND and OR node? Is this a good idea in practice? Justify your answer.